

Public Perception of AI Assistants: Anthropic vs. OpenAI

A comparative sentiment and text analysis of US App Store reviews applying lexicon-based scoring, TF-IDF weighting, and NRC emotion classification to surface brand perception differences between Claude and ChatGPT.

PROBLEM	TOOLS	OUTCOME	CODE
How do users actually feel about Claude vs ChatGPT?	R 4.6.0, tidytext, AFINN/NRC, TF-IDF, ggplot2	Claude 46% negative vs ChatGPT 32%	View Full R Script on GitHub

Summary

This analysis examines public sentiment toward two leading AI assistants, Anthropic’s Claude and OpenAI’s ChatGPT. This analysis pulls 980 US App Store reviews for Claude and ChatGPT collected May 1 to June 8, 2026, and scores them for sentiment, emotion, and theme.

Claude (Anthropic): 46% of scored reviews are negative versus 51% positive, a near-even split driven by usage limits, account restrictions, and platform bugs.

ChatGPT (OpenAI): 63% positive versus 32% negative, with broad utility language like helpful and amazing dominating the reviews.

Audience difference: Claude's most distinctive words are technical (ios, usage, banned, sonnet), while ChatGPT's include Spanish-language terms (todo, gracias, buena), pointing to a more globally distributed user base.

Four stages of analysis, AFINN sentiment scoring, NRC emotion classification, TF-IDF keyword extraction, and star rating comparison, are documented below with the full R code and results.

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01. Methodology

This analysis follows a four-stage NLP pipeline implemented entirely in R using the tidyverse ecosystem. Data was collected via the itunesr package which interfaces with the public iOS App Store review feed. No API key is required. Reviews were pulled in pages of 50, up to 10 pages per app, yielding a maximum of 500 reviews per brand.

STEP 01 Data Collection App Store reviews pulled via itunesr for Claude (App ID: 6473753684) and ChatGPT (App ID: 6448311069), US storefront, 10 pages each. 490 reviews collected per app.	STEP 02 Preprocessing Lowercasing, URL removal, punctuation stripping, whitespace normalization. Filtered to reviews with 5+ words. Stopwords removed using tidytext stop_words plus a 28-word custom domain list.
STEP 03 Sentiment Scoring AFINN lexicon for numeric valence scoring normalized by document word count. Polarity classified as Positive (>0.1), Negative (<-0.1), or Neutral. NRC lexicon for 8-emotion classification.	STEP 04 Theme Extraction Unigram frequency counts by brand. TF-IDF weighting via bind_tf_idf to surface brand-differentiating terms. Bigram analysis using unnest_tokens with ngrams token type.

R Code: Data Collection and Preprocessing

```
# Pull reviews from US App Store
pull_reviews <- function(app_id, pages = 10) {
  bind_rows(lapply(1:pages, function(p) {
    tryCatch(getReviews(app_id, "us", p), error = function(e) NULL)
  }))
}

claude_raw <- pull_reviews(6473753684) # Claude
chatgpt_raw <- pull_reviews(6448311069) # ChatGPT

# Preprocess
corpus_clean <- corpus |>
  mutate(
    text_clean = Review |>
      str_to_lower() |>
      str_remove_all("https?:/[^\s]+") |>
      str_remove_all("[^a-z0-9\s]") |>
      str_squish(),
    date_parsed = as.Date(Date),
    Rating = as.numeric(Rating),
    doc_id = row_number()
  ) |>
  filter(str_count(text_clean, "\\w+") >= 5)
```

02. Data Collection and Preprocessing

The corpus consists of 980 App Store reviews collected from May 1 to June 8 2026. After filtering reviews with fewer than 5 words, 782 documents remained for analysis. The two brands are represented at exact parity with 490 reviews each prior to filtering.

980 TOTAL RECORDS	782 AFTER CLEANING	559 AFINN SCORED	9,550 TOTAL TOKENS
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Rating distribution summary across all 782 cleaned reviews: minimum 1.0, maximum 5.0, mean 3.67, median 5.0, first quartile 1.75, third quartile 5.0. The bimodal distribution, heavily concentrated at 1 and 5 stars with almost no middle ratings, indicates that users who leave reviews are predominantly at sentiment extremes. This pattern holds for both brands but is more pronounced for Claude.

223 documents did not receive an AFINN score because none of their tokens after stopword removal matched words in the AFINN lexicon. This is a known limitation of lexicon-based approaches and is discussed further in the Limitations section.

03. Sentiment Scoring and Distribution

Sentiment was scored using the AFINN lexicon which assigns integer values from -5 (most negative) to +5 (most positive) to individual tokens. Document-level scores were normalized by word count to control for review length. Documents with normalized scores above 0.1 were classified as Positive, below -0.1 as Negative, and between the two thresholds as Neutral.

R Code - AFINN Sentiment Scoring

```
afinn_scores <- tokens |>
  inner_join(get_sentiments("afinn"), by = "word") |>
  group_by(doc_id, brand, Rating, month) |>
  summarise(
    sentiment_score = sum(value),
    word_count      = n(),
    normalized_score = sentiment_score / word_count,
    .groups = "drop"
  ) |>
  mutate(
    polarity = case_when(
      normalized_score > 0.1 ~ "Positive",
      normalized_score < -0.1 ~ "Negative",
      TRUE ~ "Neutral"
    )
  )
```

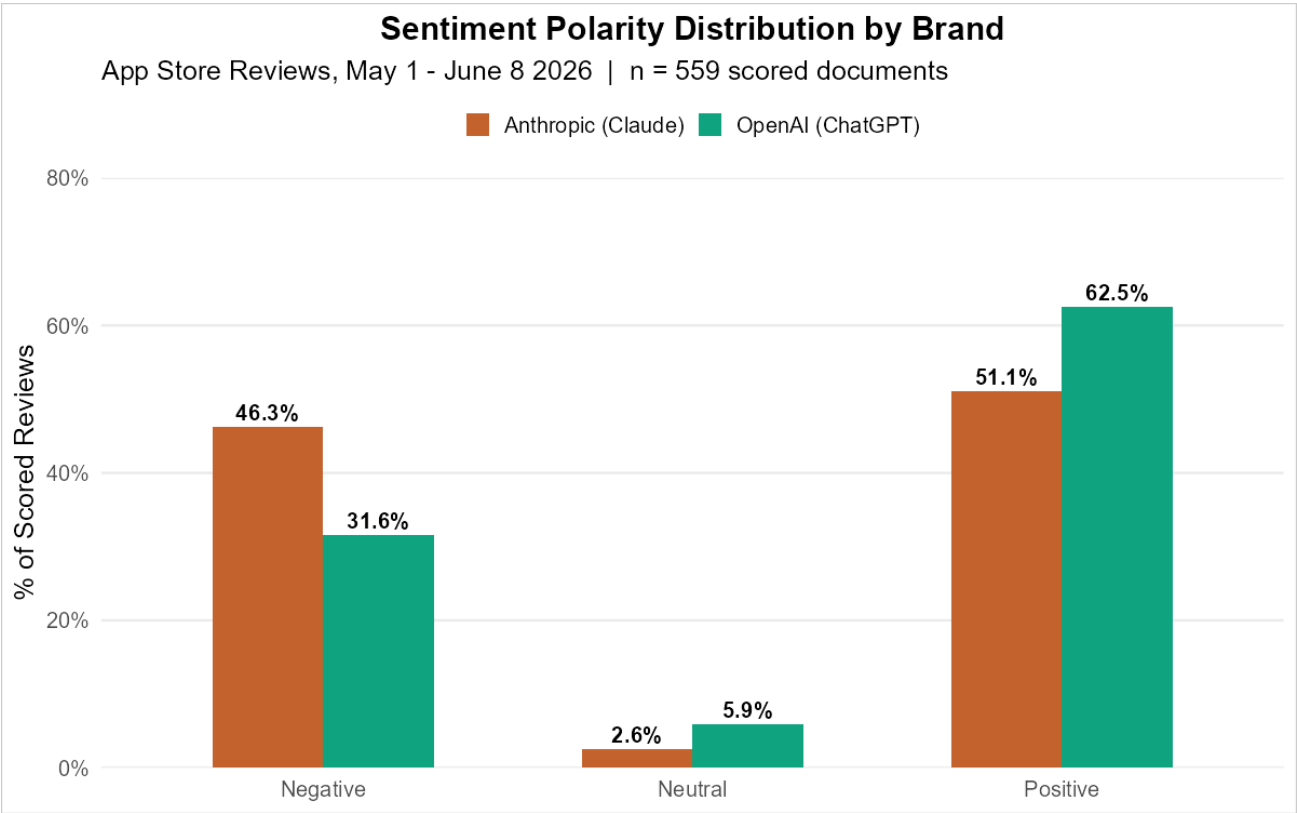


Figure 1: Sentiment polarity distribution (% of AFINN-scored reviews). Anthropic: 46% negative, 51% positive. OpenAI: 32% negative, 63% positive.

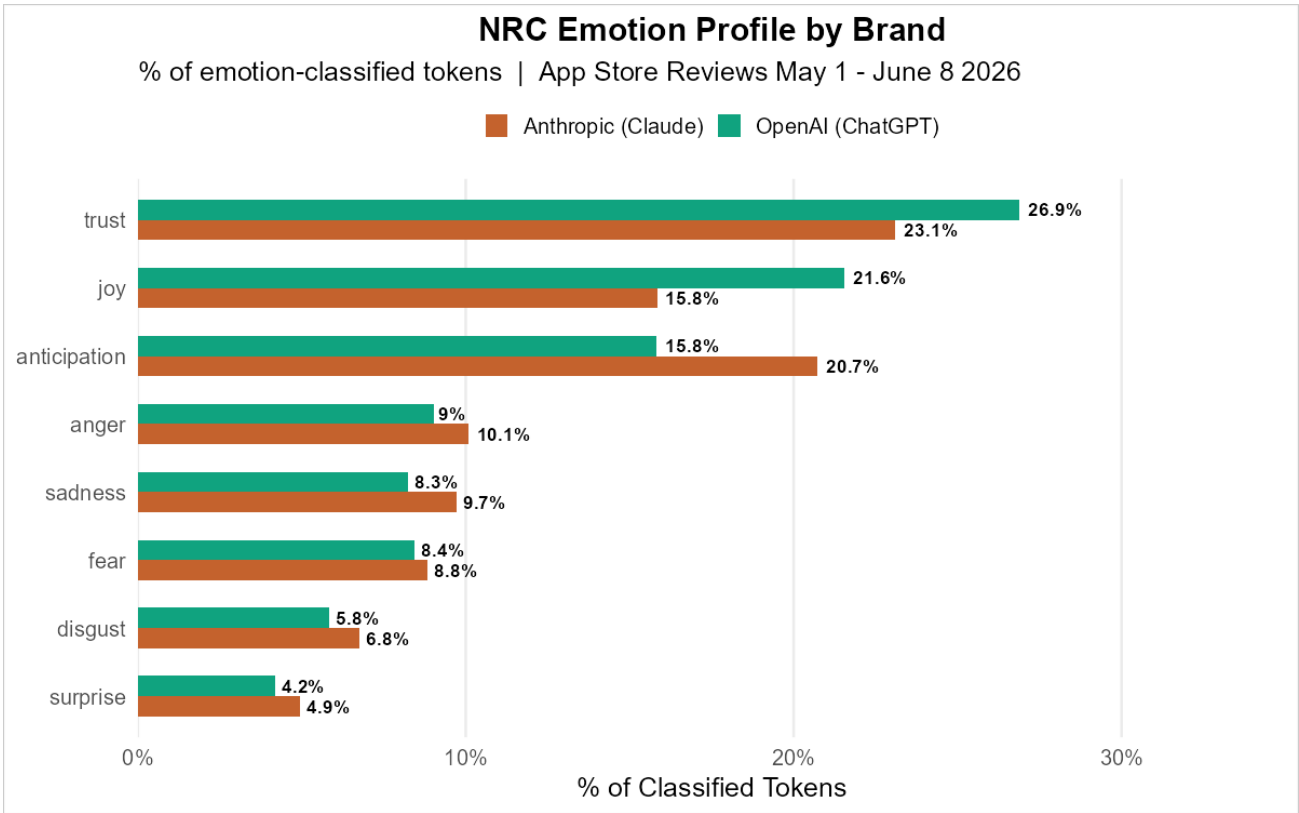


Figure 2: NRC emotion profile (% of emotion-classified tokens). Anthropic leads on anticipation, anger, sadness, fear, disgust, and surprise. OpenAI leads on trust, and joy.

KEY FINDING - ANTHROPIC (CLAUDE)

46% of scored Claude reviews are negative versus 51% positive. This near-even split is driven by a concentrated cluster of frustration around usage limits, account restrictions, and platform bugs. The high negative rate does not indicate general dissatisfaction, it reflects a specific, addressable set of product complaints from an otherwise engaged and technically sophisticated user base.

KEY FINDING - OPENAI (CHATGPT)

63% of scored ChatGPT reviews are positive with only 32% negative. The dominant positive language suggests broad utility satisfaction rather than feature-specific enthusiasm. The lower negative rate may also reflect a less technically demanding user base with fewer expectations around platform depth and model-tier access.

Representative Reviews

"Remarkable AI probably one of the better ones I've used and I've used quite a few of them." App Store - Anthropic POSITIVE	"Absolutely amazing tool. Has helped me so much in my daily life and work." App Store - OpenAI POSITIVE
"Yeah you still have a limit after spending \$20. I will stick with ChatGPT that doesn't have dumb usage limits, what a rip off." App Store - Anthropic NEGATIVE	"I have several MCP setups and it keeps telling me it doesn't have write access after I remind it. Shouldn't have done that." App Store - Anthropic NEGATIVE

04. Theme and Keyword Analysis

Unigram frequency counts and TF-IDF weighting were used to identify the most common and most brand-differentiating terms for each app. TF-IDF penalizes terms that appear with similar frequency across both brands, surfacing language that is genuinely unique to each.

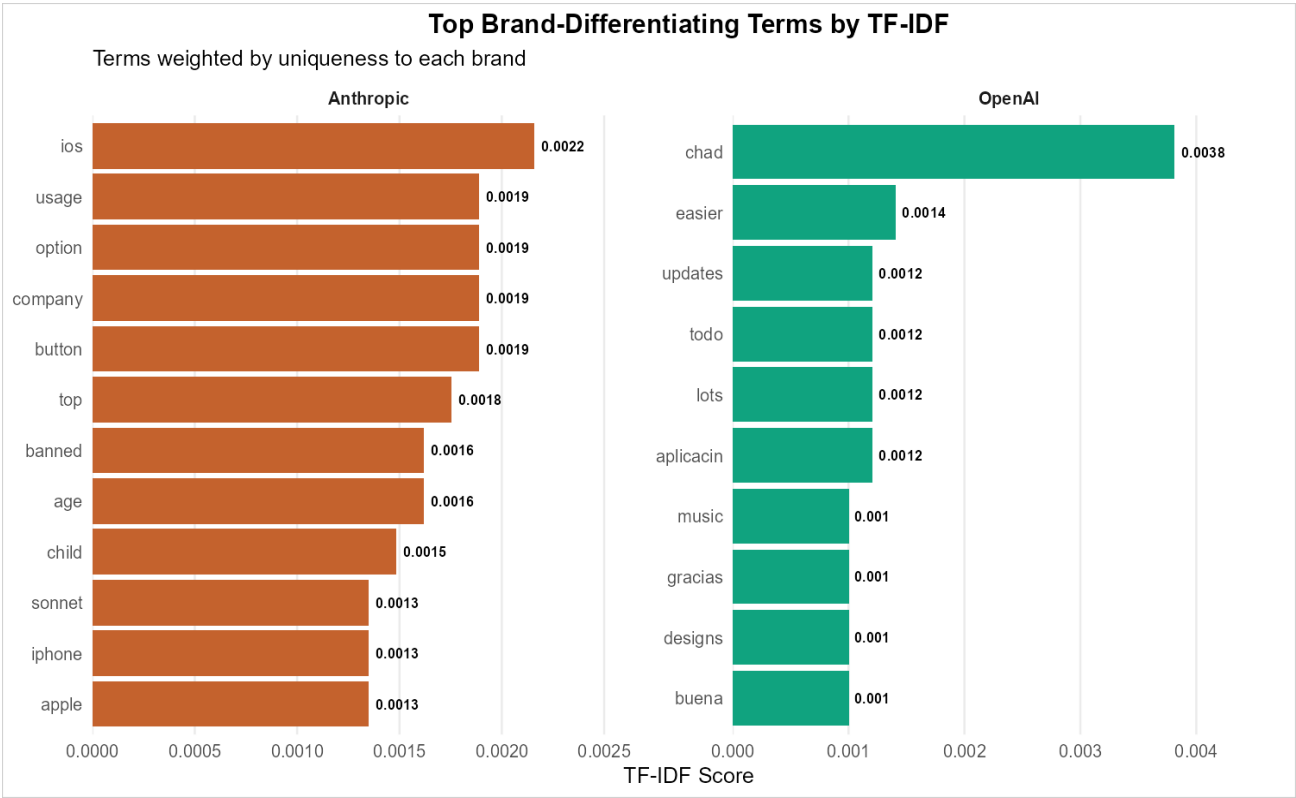


Figure 3: Top brand-differentiating terms by TF-IDF. Anthropic unique terms: ios, usage, option, company, button, top, banned, age, child, sonnet, iphone, apple. OpenAI unique terms: chad, easier, updates, todo, lots, aplicacin, music, gracias, designs, buena, confirming Spanish-language user presence.

The TF-IDF results reveal a meaningful audience difference. Anthropic's brand-unique terms are almost entirely technical or product specific: ios, usage, option, button, banned, sonnet, iphone, apple. Claude users reference specific model tiers (sonnet), specific platforms (ios). The word banned and usage appearing as a top differentiator signals that account restrictions are salient enough that users specifically write about it. The appearance of child and age as differentiating terms suggests content moderation complaints are a meaningful signal in Claude's negative reviews.

OpenAI's unique terms include significant Spanish-language content: todo (all/everything), aplicacin (application), gracias (thank you), buena (good). This indicates ChatGPT has a broader, more globally distributed user base leaving reviews in the US storefront. The top differentiating term chad, likely a proper noun or slang term, has an outsized TF-IDF score. Music and updates reflect feature-specific enthusiasm rather than frustration. Overall the OpenAI vocabulary is more casual, more global, and less technically specific than Anthropic's.

05. Star Rating Distribution

Star ratings are an objective sentiment signal that doesn't depend on any lexicon or scoring method. The distribution below confirms and reinforces the NLP findings above.

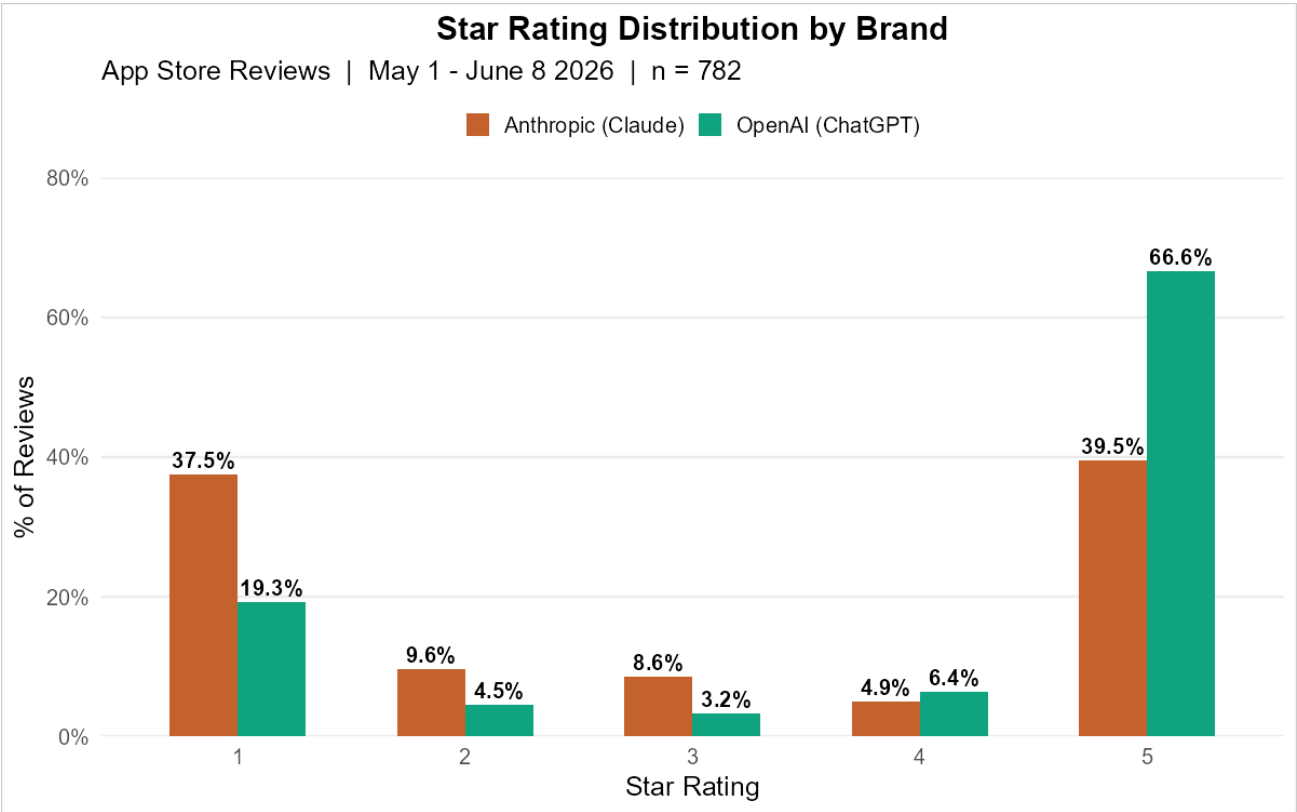


Figure 4: Star rating distribution by brand (n=782). Approximately 38% of Claude reviews are 1 star versus 19% for ChatGPT. 67% of ChatGPT reviews are 5 stars versus 40% for Claude.

Approximately 38% of Claude reviews are 1 star versus 19% for ChatGPT. Conversely, 40% of Claude reviews are 5 stars versus 67% for ChatGPT. Both distributions are bimodal with very few 2, 3, or 4 star reviews, indicating that users who leave App Store reviews are predominantly at sentiment extremes. The rating distribution is the most objective signal in this dataset, it requires no NLP and tells the same story as every other analysis: Claude users are more polarized, ChatGPT users are more consistently satisfied.

06. Key Findings and Brand Implications

Dimension	Anthropic (Claude)	OpenAI (ChatGPT)	Implication
% Negative Reviews	46%	32%	Claude generates significantly more negative reviews
% Positive Reviews	51%	63%	ChatGPT reviews skew more consistently positive
1-Star Rate	38%	19%	Claude has 2.4x the 1-star rate of ChatGPT
5-Star Rate	40%	67%	ChatGPT leads significantly on 5-star volume
Top Emotion (NRC)	Anticipation (21%)	Trust (27%)	Both anchor on positive emotions; different leading signals
Joy vs Anger Gap	Joy 16% Anger 10%	Joy 22% Anger 9%	OpenAI has a wider positive emotion margin
Primary Negative Theme	Usage limits / banned	General dissatisfaction	Claude's negatives are specific and addressable
Top Differentiating Terms	ios, usage, banned, sonnet	chad, todo, gracias, buena	Technical power users vs broad casual/global audience
Language Diversity	Primarily English	English + significant Spanish	ChatGPT has broader global reach in US storefront
Top Bigram	usage limits (n=8)	image generation (n=6)	Frustration vs feature enthusiasm as dominant signal

Strategic Implications

For Anthropic: The negative sentiment is concentrated and specific, not diffuse. Usage limits, account bans, and platform bugs account for the majority of negative reviews. This is actionable. A product team reading this analysis would know exactly where to intervene. The technical sophistication of Claude's user base (sonnet, ios, iphone, usage, coding) also suggests these users have high expectations and low tolerance for friction, which amplifies frustration when limits are hit.

For OpenAI: The positive sentiment is broad but potentially shallow. Love, amazing, and helpful are high-frequency but low-specificity positive signals. The presence of Spanish-language reviews in the US storefront suggests untapped international market depth that is not yet fully reflected in English-language marketing.

07. Limitations and Future Work

Several limitations apply to this analysis. The `itunesr` package returns only the most recent App Store reviews, limiting the dataset to May and June 2026. A longer historical window would enable temporal trend analysis. The 223 documents that did not receive AFINN scores (29% of the cleaned corpus) represent a meaningful coverage gap. Reviews using slang, informal contractions, or non-English language are underrepresented in the scored set.

Lexicon-based methods misclassify sarcasm and negation at higher rates than transformer-based models. A review saying 'not helpful at all' would have 'helpful' scored as positive despite the negation. Future work could apply a fine-tuned RoBERTa or BERT classifier to improve accuracy on informal App Store text. Entity-level sentiment attribution, separating opinions about pricing from opinions about response quality, would also add analytical depth.

R Session Info

```
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64

Attached packages:
itunesr      tidytext    syuzhet     ggplot2
tidyverse    tidyr       stringr     lubridate
scales       textdata    wordcloud   reshape2
```

All data collected from the US iOS App Store via the `itunesr` R package (github.com/amrrs/itunesr). Raw data saved to `app_store_corpus.csv`. Analysis objects saved to `sentiment_analysis.RData`. Full reproducible script: <https://github.com/shaniahe2002-cyber/sentiment-analysis-anthropic-openai/tree/main>. AFINN and NRC lexicons accessed via the `textdata` R package.

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